

Types of Threats (Attacks)

Security attacks are classified into two major categories

1. Passive Attacks
2. Active Attacks

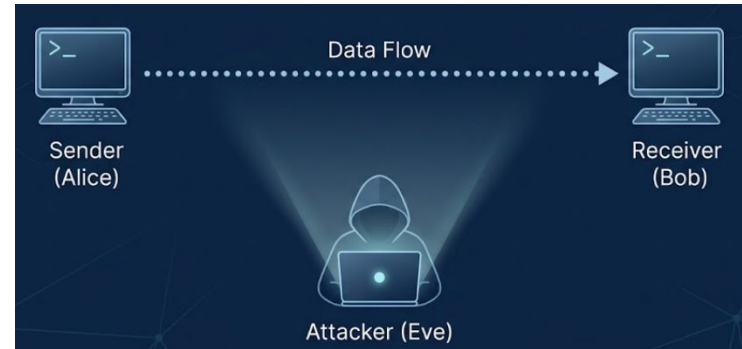
Understanding these threats helps in designing secure systems

Passive Attacks

- Passive attacks monitor information without modifying it
- The attacker gathers information secretly
- Information remains unchanged
- Focus is on data collection

Characteristics of Passive Attacks

- Difficult to detect
- No modification of data
- Communication continues normally
- Confidentiality is compromised



Types of Passive Attacks

- Release of Message Contents
- Traffic Analysis

Release of Message Contents

- Attacker reads confidential information
- Example: Reading email messages without authorization
- Sensitive business or personal information may be exposed



Traffic Analysis

- Attacker observes communication patterns
- Example: Monitoring network traffic between departments
- Even encrypted communication may reveal useful information

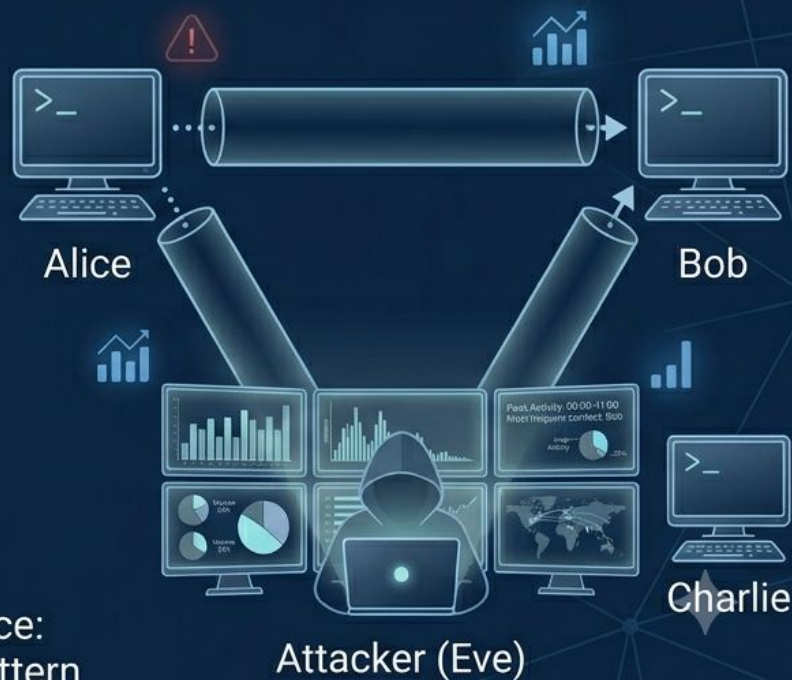


RELEASE OF MESSAGE CONTENTS



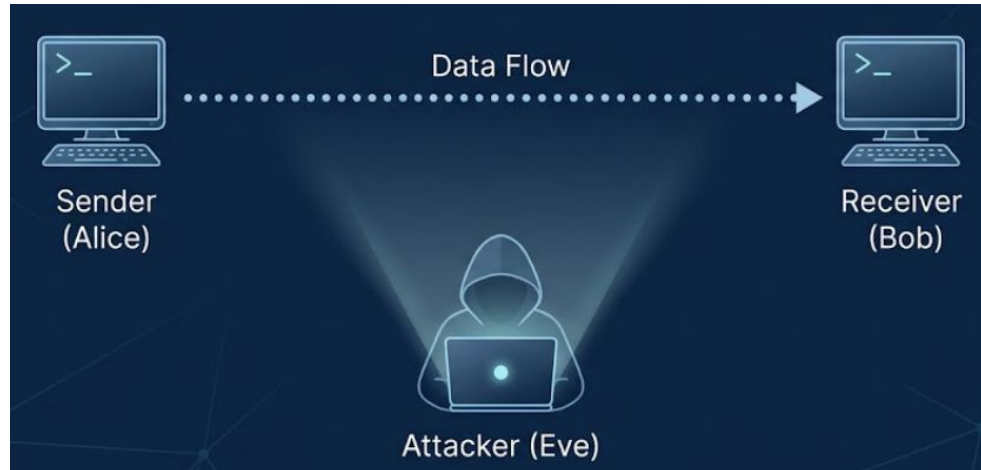
Key Difference:
Content vs. Pattern

TRAFFIC ANALYSIS



Passive Attack Diagram

- Sender -----> Receiver
- Attacker observes communication only
- No data modification occurs



Active Attacks

- Active attacks involve modification, interruption, or destruction of data
- System operations are affected
- Integrity and availability may be compromised



Characteristics of Active Attacks

- Easier to detect than passive attacks
- Data may be modified
- Services may be disrupted
- System resources can be misused

Types of Active Attacks

- Masquerade Attack
- Replay Attack
- Modification Attack
- Denial of Service Attack

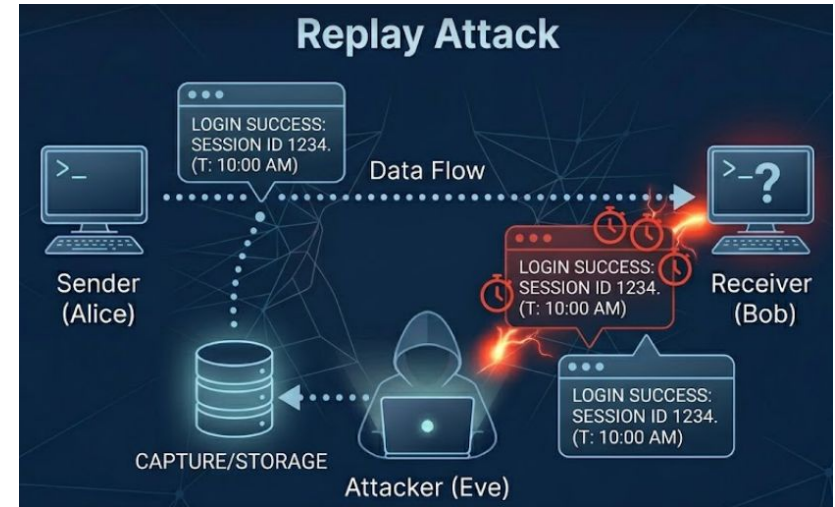
Masquerade Attack

- Attacker pretends to be an authorized user
- Example: Using stolen login credentials
- Can lead to unauthorized access



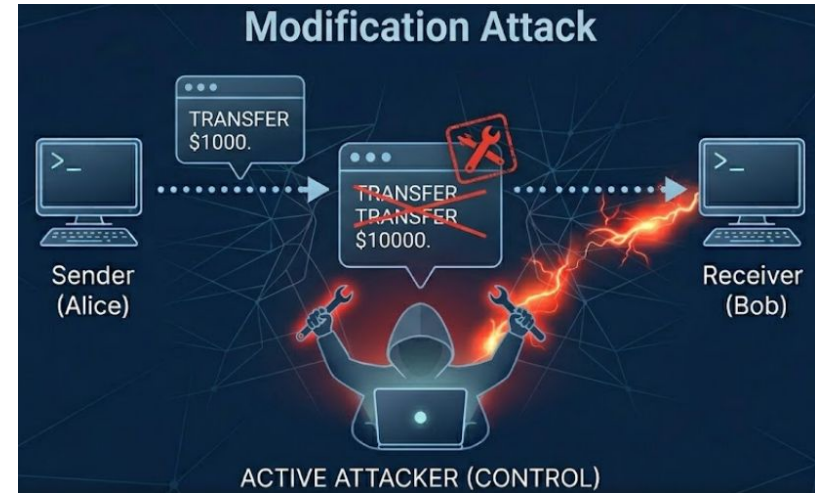
Replay Attack

- Captured information is retransmitted
- Example: Resending authentication messages
- Can bypass authentication mechanisms



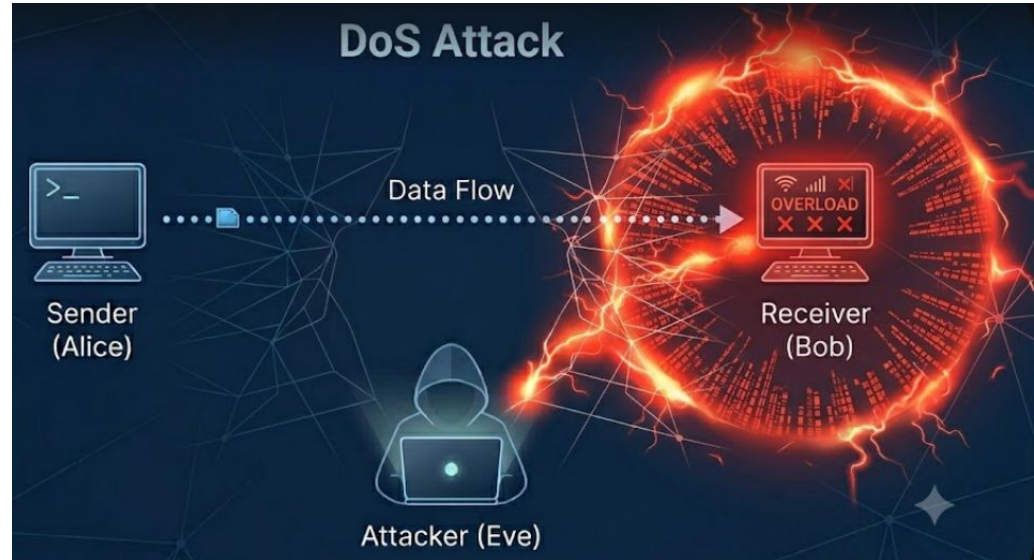
Modification Attack

- Data is altered during transmission
- Example: Changing transaction amounts in online banking
- Integrity of information is compromised



Denial of Service Attack

- Services become unavailable to legitimate users
- Example: Flooding a website with excessive requests
- Affects system availability



Active Attack Diagram

- Sender -----> Attacker -----> Receiver
- Data is modified before reaching receiver



Additional Modern Threats

- Malware
- Phishing
- Social Engineering
- Insider Threat

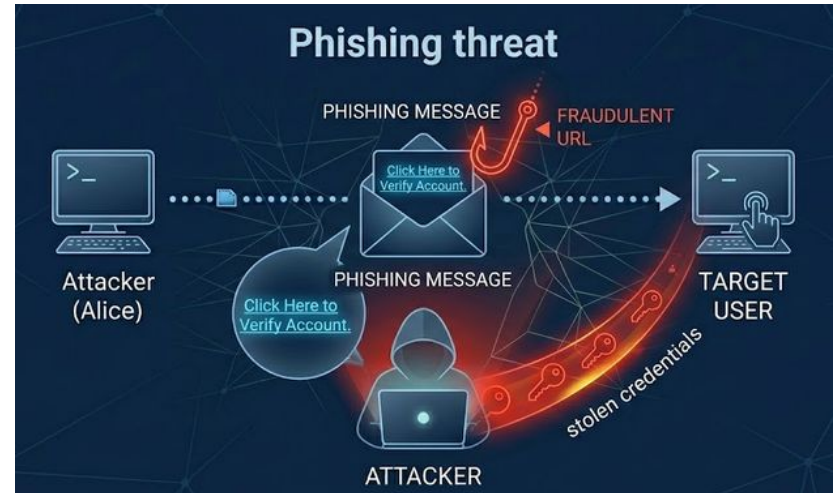
Malware

- Malicious software designed to damage systems
- Types include Virus, Worm, Trojan Horse, Spyware and Ransomware
- Can steal, destroy or encrypt data



Phishing

- Fraudulent attempts to obtain sensitive information
- Example: Fake banking emails requesting login credentials
- Often uses email, messages or fake websites



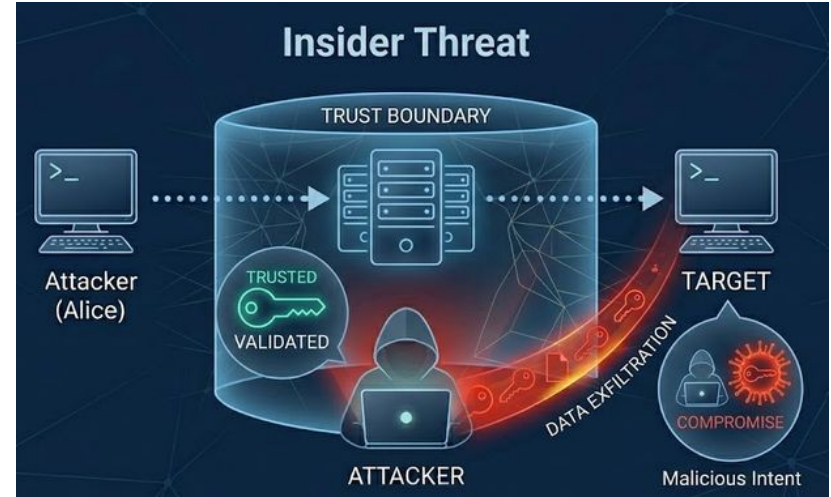
Social Engineering

- Manipulating people into revealing confidential information
- Example: Pretending to be technical support
- Targets human behavior rather than technology



Insider Threat

- Threat originating from authorized users
- Example: Employee leaking confidential documents
- Can be intentional or accidental



Summary

- Passive attacks observe information
- Active attacks modify or disrupt information
- Modern threats include malware, phishing, social engineering and insider threats
- Security controls are essential to protect confidentiality, integrity and availability

Thank you

Any Questions?